

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EC-8003 (3) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Digital Control System
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Determine the Z-transform of the following signals
 - i) $x(n) = (1+n)u(n)$
 - ii) $x(n) = (a^n + a^{-n})u(n)$, 'a' real
 - iii) $x(n) = (na^n \sin \omega_0 n)u(n)$
 - iv) $x(n) = \left(\frac{1}{2}\right)^n [u(n) - u(n-10)]$
2.
 - a) What is pulse transfer function also discuss sampled signal flow graph.
 - b) Describe Lyapunov stability analysis in detail.
3. State and prove sampling theorem. How the original signal is reconstructed from sampled signal? What is aliasing effect and how can it be eliminated?
4.
 - a) How is the mapping done between the S-plane and the Z-plane?
 - b) Discuss about the stability analysis of closed loop system in Z-plane.
5.
 - a) Explain about the Jury stability test.
 - b) With suitable example explain design based on Root locus method.
6.
 - a) Discuss about state space representation of discrete time system.
 - b) Discuss about discretization of continuous time state space equation.
7.
 - a) Discuss about Controllability and Observability.
 - b) Discuss about useful transformation in state space analysis and design.
8. Write short notes on any two of the following:
 - a) Any five properties of Z-transform
 - b) Limitations of Z-transform methods
 - c) State Observers
